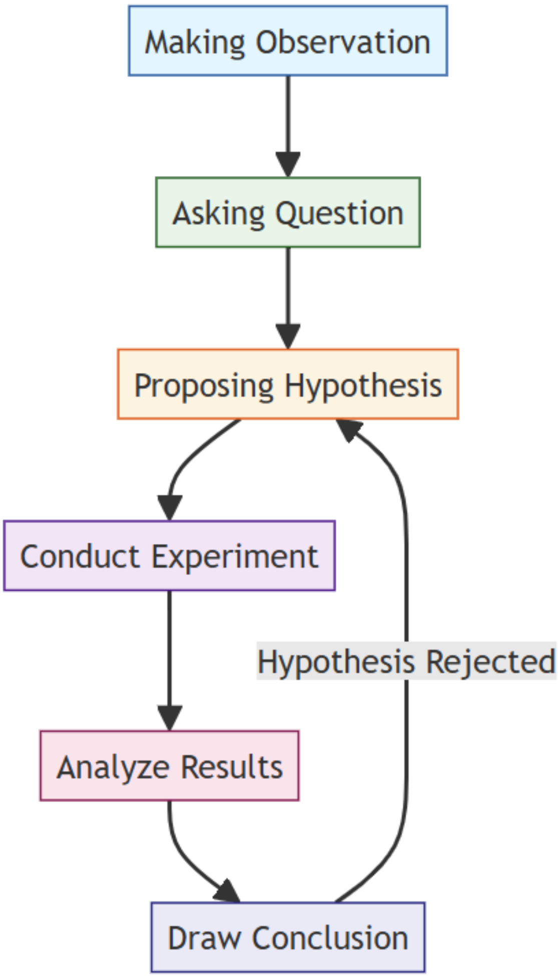
Chapter 1 – Introduction to Science

Identify the steps of scientific investigation

Steps of Scientific investigation 科學調查



1. Making Observation (觀察): Identify phenomena 現象
2. Asking a Question (問題): Ask a question related to the observation
3. Proposing 提出 a Hypothesis (假設): Form an educated guess on the question
4. Doing Experiment (實驗): Design and perform experiments and collect data
5. Doing Analysis (分析): Analyze the results. E.g. plotting graphs
6. Drawing a Conclusion (結論): Draw conclusions based on the data.

Example : Does the Weight of McDonald's Burgers Decrease Over Time?

Fill in the blank with one of the following words:

Observation (觀察), Question (問題), Hypothesis (假設), Experiment (實驗),

Analysis (分析), Conclusion (結論)

1. Making ____________________:

You notice that McDonald's burgers seem lighter than they used to be.

2. Asking a ____________________:

Does the weight of McDonald's burgers decrease over time?

3. Proposing a ____________________:

The weight of McDonald's burgers decreases over time.

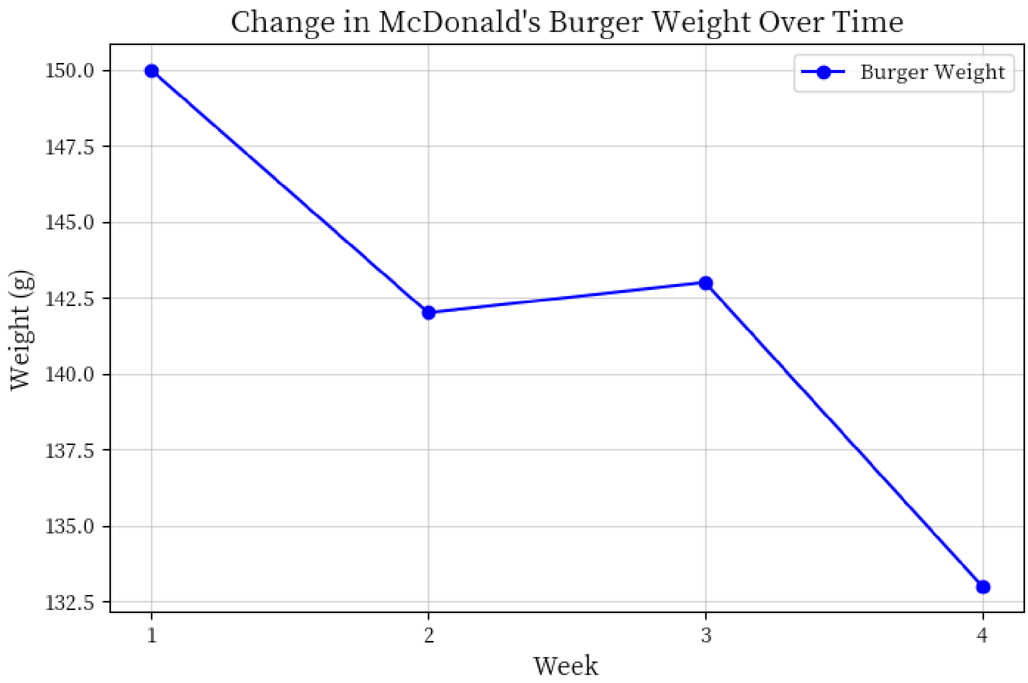
4. Doing ____________________:

- **Procedure:**
  i. Buy a McDonald's burger once every week.
  ii. Measure and record the weight of each burger using the kitchen scale 廚房秤.
  iii. Repeat this process over 4 weeks.

5. Doing ____________________:

- Create a table to record the weight measurements for each week.

Week	Weight (g)
1	150
2	142
3	143
4	133



the weight of McDonald’s burger shows a **trend** 趨勢 of [increase / decrease] over 4 weeks.

6. Drawing a ______________________:

From the results, the hypothesis 假設 is [supported / not supported].

➔ The weight of McDonald’s burgers [does / does not] decreased over weeks.

➔ So we [need / do not need] to propose another hypothesis and do another experiment.

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Scenario exploration

Try to identify correspond the different part of the essay to the steps of the scientific investigation

Scenario 1: Plant Growth and Light（植物生長與光照）

	Stages of scientific investigation
You notice that the plants near the window grow faster than those far from the window. 你發現家中的植物在靠近窗戶的地方長得比遠離窗戶的地方更快。	
You start to wonder why the plants near the window grow faster? 你開始思考，為什麼靠近窗戶的植物長得比較快？	
You think, maybe more sunlight allows plants to grow faster. 你心裡想，也許是因為陽光較多讓植物長得更快。	
You decide to place two identical plants, one near the window and one inside the room, giving both the same water and fertilizer, and record their heights each week. 你決定把兩盆相同的植物分別放在窗戶旁和房間內，並給它們相同的水和肥料，每週記錄它們的高度。	
You organize the weekly plant heights into a table and plot a growth curve. 你將每週的植物高度整理成表格，並畫出生長曲線圖。	
You find that the plant near the window grows noticeably taller, so you make a conclusion based on the data. 你發現窗戶旁的植物長得明顯更高，於是根據數據作出總結。	

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Scenario 2: Effect of Temperature on Salt Solubility

（溫度對鹽溶解度的影響）

	Stages of scientific investigation
You notice that salt dissolves faster in hot water. 你注意到熱水能更快地溶解鹽。 You start to wonder, does temperature affect the solubility of salt? 你開始思考，溫度是否會影響鹽的溶解度？ You guess that the higher the water temperature, the more salt can dissolve. 你猜想水溫越高，能溶解的鹽會越多。 You heat equal amounts of water to different temperatures, add salt until it no longer dissolves, and record the amount dissolved at each temperature. 你把等量的水分別加熱到不同溫度，然後加入鹽直到不再溶解，並記錄每種溫度下溶解的鹽的質量。 You create tables and graphs to compare the amount of salt dissolved at different temperatures. 你把不同溫度下溶解的鹽的質量製成表格和圖表進行比較。 You see that more salt dissolves at higher temperatures, and then make a conclusion based on these results. 你看到高溫下溶解的鹽較多，於是根據這些結果作出總結。	

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Scenario 3: Warm up Rate of Drinks（飲品的回溫速度）

A . You start to wonder why cold drinks in metal cups warm up faster than those in plastic cups.
你開始思考，為什麼金屬杯的冷飲會比塑膠杯的冷飲更快回溫？

B . You notice that cold drinks in metal cups warm up faster than those in plastic cups.
你發現用金屬杯盛載的冷飲比用塑膠杯的冷飲更快變溫。

C . You find that the drink in the metal cup warms up faster, and then make a conclusion based on the results.
你發現金屬杯中的飲品升溫更快，於是根據結果作出總結。

D . You organize the measured temperatures into a table and line graph to observe the changes in both cups.
你把每次測量的溫度整理成表格和線圖，觀察兩種杯子的變化。

E .You think, maybe metal conducts heat faster than plastic, so the drink warms up more quickly.
你心裡想，可能是因為金屬導熱比塑膠快，所以飲品會更快回溫。

F .You pour cold drinks of the same temperature into a metal cup and a plastic cup, and measure the temperature every five minutes.
你將相同溫度的冷飲分別倒入金屬杯和塑膠杯，每隔五分鐘測量一次飲品的溫度。

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